THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 18 m s⁻¹	900 ÷ 50
2	**(d)** 30 m s⁻¹	108 × 5/18
3	**(b)** −3 m s⁻²	(0 − 15) ÷ 5
4	**(c)** 25 m	s = ut + ½at²
5	**(b)** 90 m	v² = u² + 2as
6	**(c)** 12 N	F = ma
7	**(d)** 4 m s⁻²	a = F/m
8	**(b)** 600 N	W = mg
9	**(c)** 6 N	add if same direction; subtract if opposite
10	**(d)** 6 m s⁻¹	a = F/m, v = at

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Velocity (rate of change of position) (2).	2
12	Displacement (1); rectangle: velocity × time (1).	1+1
13	Friction is very small, so the foot cannot push the ground backwards and the ground cannot push forward enough (2).	2
14	Net force is zero, so by the first law it continues at constant velocity if already moving (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = rectangle ut + triangle ½ × t × (v − u) (1.5); (v − u) = at, so s = ut + ½at² (1.5).	1½+1½
16	Bullet: 2/0.1 = 20 m s⁻² (1.5); gun: 2/5 = 0.4 m s⁻² in the opposite direction (1.5).	1½+1½
17	a = 20/50 = 0.4 m s⁻² (1.5); s = ½ × 0.4 × 50² = 500 m (1.5).	1½+1½
	OR: 0 = 400 − 10s ⇒ s = 40 m (2); t = 20/5 = 4 s (1).	2+1
18	The Earth's mass is huge, so its acceleration a = F/m is negligibly small (3).	3
	OR: The stretched balloon pushes the air out backwards (1.5); the air exerts an equal force on the balloon forward, so it moves (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 40 m (1). (ii) No, 10 m short (1). (iii) 90 m (2). (iv) Stopping distance rises as the square of speed, so speed limits matter (1).	1+1+2+1
	OR: (i) Motion along a circle at constant speed (1). (ii) Direction of velocity changes (1). (iii) On removing the ring, the marble moves along a straight tangent (2). (iv) Real motions seldom have exactly constant speed and a perfect circle (1).	1+1+2+1
20	(i) Force opposing relative motion between surfaces in contact (1). (ii) Pull a block with a spring balance on surfaces; reading when it just moves is the friction (2). (iii) Foot pushes back, friction pushes forward (2).	1+2+2
	OR: (i) 2.5 m s⁻² (1). (ii) 2000 N (1). (iii) 125 m (2). (iv) 500 m (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the shortest straight-line change in position with a direction	vector quantity
2	**(d)** velocity	rate of change of position
3	**(a)** constant velocity, zero acceleration	slope is zero
4	**(d)** zero	returns to start
5	**(b)** a straight inclined line on a velocity–time graph	constant slope of v–t
6	**(c)** inertia of rest	the body tries to remain at rest
7	**(d)** act on different objects	third law
8	**(d)** increase friction	grip
9	**(c)** smaller acceleration	$a \propto 1/m$
10	**(d)** 2.5 m s^{-2} east	(8 − 3)/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$v = u + at$ (½); $s = ut + \frac{1}{2}at^2$ (1); $v^2 = u^2 + 2as$ (½).	2
12	$v = 0 + 3 \times 4 = 12$ m s^{-1} (2).	2
13	Net force = 10 − 6 = 4 N in the direction of the larger force (2).	2
14	It increases the time to stop the ball, so the acceleration and force are smaller (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stopping distance $\propto u^2$ ($v^2 = u^2 + 2as$); doubling speed makes it four times (3).	3
16	Fire its engine in the direction of motion; exhaust exerts an opposite force on the craft (3).	3
17	Motion in a circle at constant speed, e.g. a fan blade tip (1); direction of velocity keeps changing (1), so it is accelerated (1).	1+1+1
	OR: Distance = 88 m (1); displacement = 0 (1); speed = 88/22 = 4 m s^{-1} (1).	1+1+1
18	Total mass 5 kg, $a = 10/5 = 2$ m s^{-2} (1.5); $T = 2 \times 2 = 4$ N (1.5).	1½+1½
	OR: Mass: amount of matter, constant, kg, scalar (1.5); weight: gravitational force on the body, depends on g, N, vector (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 5 × 44 = 220 m (1). (ii) 0 (1). (iii) 220/110 = 2 m s^{-1} (2). (iv) 0 (1).	1+1+2+1
	OR: (i) 9.8 m s^{-2} each (1). (ii) Yes (1). (iii) Acceleration due to gravity (g) (1). (iv) ½ × 9.8 × 9 = 44.1 m (2).	1+1+1+2
20	(i) Treat both and the string as one object; internal tension is ignored, $a = F/(m_1 + m_2)$ (2). (ii) 2 m s^{-2} (1.5). (iii) $T = m_2 a = 6$ N when box 1 is pulled (1.5).	2+1½+1½
	OR: (i) Balanced: net force zero; unbalanced: non-zero (2). (ii) 100 N towards the first team (2). (iii) First team, as net force is in its direction (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 15 m s^{-1}	1500 ÷ 100
2	**(d)** 4 m s^{-2}	(20 − 0) ÷ 5
3	**(c)** 18 m s^{-1}	v = u + at
4	**(d)** 56 m	$s = ut + \frac{1}{2}at^2$
5	**(c)** 25 m	$v^2 = u^2 + 2as$
6	**(b)** 100 N	F = ma
7	**(d)** 1.5 m s^{-2}	a = F/m
8	**(c)** 4 N	add if same direction; subtract if opposite
9	**(a)** 20 kg	m = F/a
10	**(d)** newton	kg m s^{-2}

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The direction of velocity changes continuously, so velocity changes (2).	2
12	Tangent to the circle at the point (2).	2
13	W = mg = 30 × 10 = 300 N (2).	2
14	Forces which are equal and opposite on the same object so the net force is zero (1); a book at rest on a table (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	h = ½ × 10 × 9 = 45 m (2); v = 30 m s^{-1} (1).	2+1
16	a = 2 m s^{-2}, F = 3000 N forward (2); on stopping, F = −3000 N (1).	2+1
17	B is faster (steeper) (1); A = 50 m, B = 100 m (2).	1+2
	OR: No; if it turns a bend, the direction of velocity changes and so it accelerates (3).	3
18	Net = (95 − 5) × 200 = 18 000 N (3).	3
	OR: Every force is an interaction between two objects, giving equal and opposite forces on each (1); hammer–nail and Earth–apple (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Scalar: distance, speed; vector: displacement, velocity (acceleration) (2). (ii) Path length vs shortest change in position; scalar vs vector; never zero for moving vs can be zero; ≥ displacement (3).	2+3
	OR: (i) 25 m s^{-1} (1). (ii) 25/50 = 0.5 m s^{-2} (1). (iii) s = 25 × 50/2 = 625 m (2). (iv) First 25 s: 25 × 25 − ½ × 0.5 × 25^2 = 468.75 m, i.e. 3/4 of the total (1).	1+1+2+1
20	(i) Weight downwards, her upward force (1). (ii) 294 N (1). (iii) 294 N upwards (1). (iv) Zero (2).	1+1+1+2
	OR: (i) Mass is fixed, weight depends on g (2). (ii) 600 N (1.5). (iii) 96 N (1.5).	2+1½+1½

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** zero	returns to start
2	**(a)** acceleration	rate of change of velocity
3	**(d)** $v = s/t^2$	not valid in general
4	**(d)** 9.8 m s^{-2}	g
5	**(c)** $2\pi r/T$	circumference ÷ time
6	**(c)** opposite to its motion	opposes relative motion
7	**(b)** Newton's third law	equal and opposite force
8	**(c)** force of friction	reaction to backward push
9	**(c)** keeps moving indefinitely if no friction acts	basis of the first law
10	**(d)** 10 kg m s^{-1}	p = mv

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Uniform: a train at constant speed on a straight track; non-uniform: a car speeding up from a traffic signal (1+1).	1+1
12	$v = gt = 10 \times 2 = 20$ m s^{-1} (2).	2
13	The exhaust gases pushed down exert an equal upward force on the rocket, larger than its weight (2).	2
14	Forces are equal and opposite but the gun is much heavier, so its acceleration is much less (a = F/m) (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Combined speed = 14 yojanas/day (1.5); time = 210/14 = 15 days (1.5).	1½+1½
16	$a = 8/4 = 2$ m s^{-2} (1.5); $F = 4 \times 2 = 8$ N (1.5).	1½+1½
17	20/2 = 10 m s^{-1}, 40/2 = 20 m s^{-1} (2); unequal distances in equal times, so non-uniform (1).	2+1
	OR: $0 = 784 + 2a(98) \Rightarrow a = -4$ m s^{-2} (2); $t = 28/4 = 7$ s (1).	2+1
18	Treads increase friction/grip (1.5); water reduces friction so tyres slip and stopping distance rises (1.5).	1½+1½
	OR: Moving objects would never stop, we could not walk, knots and nails would not hold, vehicles could not grip (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) B (1). (ii) A 60 m, B 120 m (2). (iii) Gap grows 10 m per second, so 10 s (1). (iv) The car is at rest (1).	1+2+1+1
	OR: (i) Straight-line motion with constant acceleration (1). (ii) $0 = 784 + 2a(98)$, $a = -4$ m s^{-2} (2). (iii) $t = 28/4 = 7$ s (1). (iv) It acts opposite to the velocity (1).	1+2+1+1
20	(i) Rate of change of momentum is proportional to net force (1). (ii) Product of mass and velocity (1). (iii) $\Delta p = 10$, $F = 10/0.5 = 20$ N (3).	1+1+3
	OR: (i) First: lurch in a bus; second: F = ma for a car; third: rocket launch (3). (ii) First law (inertia) – the body continues moving forward (2).	3+2

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